

Bilal Delais

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PROFILE

Final-year engineering student at IMT Mines Alès, specialising in Artificial Intelligence & Data Science (2IA-IASD). Currently completing a final-year internship at **ArianeGroup** on deep learning-based anomaly detection for rocket engine test benches (Vulcain, Vinci, Prometheus). Seeking a permanent position from October 2026, with a priority focus on aerospace, defence, or advanced industry sectors.

PROFESSIONAL EXPERIENCE

AI / Data Science Engineer — Propulsion Systems Mar. 2026 – present · Vernon, France

ArianeGroup — Final-year internship (targeting permanent hire)

- Design and training of a multivariate anomaly detection pipeline on time series — **Transformer-GAN**, **LSTM Autoencoder**, and **Transformer for Anomaly Detection** architectures, parallel implementations in PyTorch / TensorFlow.
- Synthetic anomaly generation to address the scarcity of real-world cases; architecture benchmarking and hyperparameter tuning via **Optuna**.
- Automated generation of test exploitation pre-reports (full curves + zooms on critical areas) for test engineers.

Machine Learning Developer (volunteer) Jun. 2025 – Mar. 2026 · Paris (remote)

SpringLane — Retail startup

- Development of a stock prediction algorithm for an omnichannel retailer, integrated into the store sales analysis strategy.
- Optimisation of automated replenishment based on the predictive model.

Operational Reservist — 503rd Train Regiment Jan. 2023 – Jan. 2025 · Nîmes-Garons

French Army — Operational Reserve

- Engaged alongside studies (logistical support) — experienced in security-sensitive and confidential environments, relevant to dual-use and defence contexts.

R&D Internship — Propulsion Systems Design Aug. – Dec. 2024 · Talence, France

Agena Space SAS — Aerospace propulsion and tank solutions

- 3D modelling and modification of propulsion components and systems (SolidWorks); production of technical drawings and assembly files.
- Managed the test bench manufacturing and assembly campaign; integration of propulsion sub-systems.

SELECTED TECHNICAL PROJECTS

- Deep Learning Correction of the SGP4 Orbital Propagator** — residual error learning on TLE data (ISS, Hubble); MLP / LSTM / Transformer benchmark, with the Transformer encoder achieving a stable and physically coherent position error reduction.
- NoduLoCC 2026 — Medical AI Challenge** — pulmonary nodule detection on chest X-rays (RadDino architecture, multi-task classification + localisation head), trained on LIDC / NIH datasets.
- Self-supervised Learning & Domain Adaptation** — CLIP, SimCLR, DANN; object detection with DETR on event cameras (DSEC-MOD).

EDUCATION

Engineering Degree — AI & Data Science (2IA-IASD) 2023 – 2026

IMT Mines Alès · Alès, France — Deep learning (CNN, RNN, Transformers, GANs), computer vision, NLP, time series.

Preparatory Classes for Engineering Schools (CPGE) — PSI Track 2020 – 2022

Lycée Gustave Eiffel · Bordeaux, France

TECHNICAL SKILLS

ML / Deep learning : PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, Optuna

Data & languages : Pandas, NumPy, SQL, Matplotlib — Python, C/C++, Java

Tools & MLOps : Git, GitLab CI, Jupyter, Linux | **CAD** : SolidWorks, 3DEXperience, Catia, Matlab, Ansys

LANGUAGES

French — native · **English** — C1

INTERESTS

Basketball, running, weightlifting, swimming · F1 motorsport · personal finance